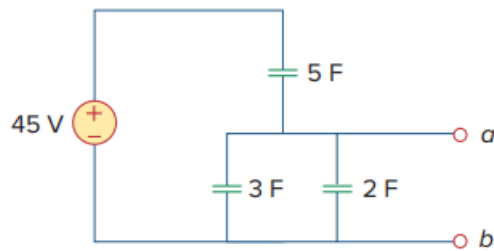


Problem

Obtain the Thevenin equivalent at the terminals, a - b , of the circuit shown in Fig. 6.65. Please note that Thevenin equivalent circuits do not generally exist for circuits involving capacitors and resistors. This is a special case where the Thevenin equivalent circuit does exist.

Fig. 6.65:



Step-by-step solution

Step 1 of 5

Draw the following circuit diagram:

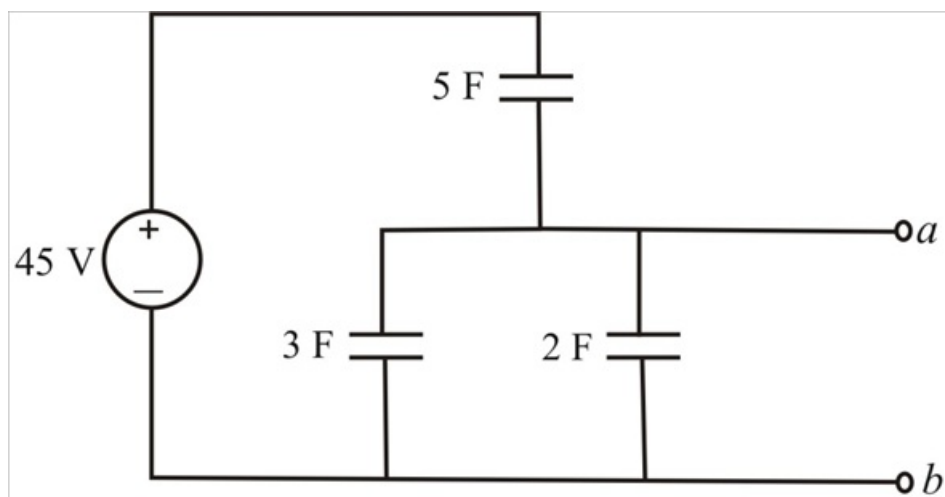


Figure 1

Step 2 of 5

Find the Thevenin equivalent across the terminals a,b :

Find the Thevenin equivalent capacitance across the terminals a,b :

Draw the following circuit diagram to find Thevenin capacitance across the terminals a,b :

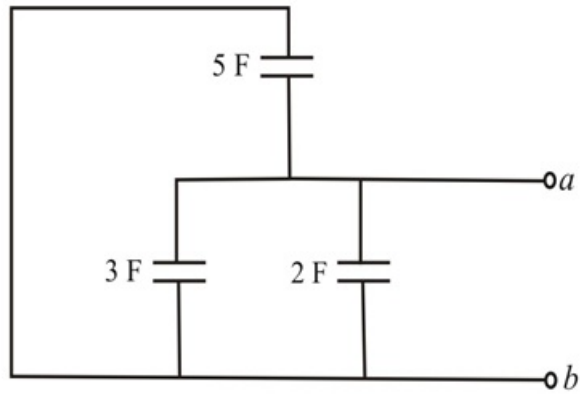


Figure 2

Step 3 of 5

Calculate the value of Thevenin capacitance across the terminals a,b :

$$\begin{aligned} C_{th} &= 3 \text{ F} + 2 \text{ F} + 5 \text{ F} \\ &= 10 \text{ F} \end{aligned}$$

Therefore, the value of Thevenin capacitance across the terminals a,b is 10 F.

Step 4 of 5

Find the Thevenin voltage across the terminals a,b :

Clearly from Figure 1, write the expression for the Thevenin voltage across the terminals a,b .

$$\begin{aligned} v_{th} &= \frac{5}{5+5} 45 \text{ V} \\ &= 22.5 \text{ V} \end{aligned}$$

Therefore, the value of Thevenin voltage across the terminals a,b is 22.5 V.

Step 5 of 5

Draw the following circuit diagram:

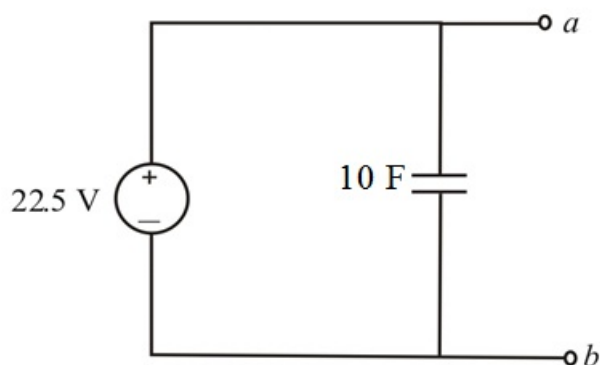


Figure 2

Hence, the required Thevenin equivalent circuit is as shown in Figure 2.